

Semester Test 3

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Electricity and Electronics EBN 111

30 June 2021

Test Information

Maximum marks:	46	Full marks:	45
Duration of test:	100 min	Open/Closed book:	Open
Total number of pages (this page included)		17	

Lecturers: Dr D. le Roux, Dr S. Smith, Dr X. Ye**Assistant Lecturers:** L. Kokot, M. Maritz, J. Thiselton, M. Trivedi, D. Wepener**Important**

1. The examination regulations of the University of Pretoria apply.
2. Write your answers on the Excel spreadsheet found on the AMS system. The spreadsheet contains unique parameter values for each student. Therefore, you must download the Excel spreadsheet from your own AMS account. Follow the instructions on the AMS spreadsheet carefully.
3. Final answers must be written as real numbers and not as fractions. Use at least 5 significant figures to write numbers. Use decimal points and NOT decimal commas.
4. Answers must include units where applicable. (Refer to the EECE Rules for Electronic Assessments of tests and examinations for guidelines on how to enter units. Also refer to the example list of units on the Excel spreadsheet.)
5. **By submitting your semester test, you declare the following:**

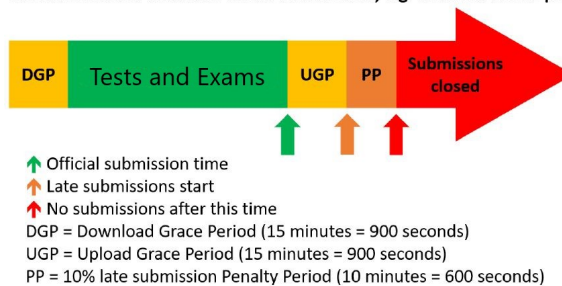
The University of Pretoria commits itself to produce academic work of integrity. I affirm that I am aware of and have read the Rules and Policies of the University, more specifically the Disciplinary Procedure and the Tests and Examinations Rules, which prohibit any unethical, dishonest or improper conduct during tests, examinations and/or any other forms of assessment. I am aware that no student or any other person may assist or attempt to assist another student, or obtain help, or attempt to obtain help from another student or any other person during tests, examinations and/or any other forms of assessment.

Online assessment submission rules and time-line

The following rules and time-line apply for all tests for EBN111/122 as illustrated by the figure below:

1. The Download Grace Period (DGP) is a 15 minute period before the official start of the test for students to download all documents related to the test and read all the necessary instructions.
2. The start of the green section after the DGP indicates the official start of the test. Students must work through the test at an average rate of at least 2 minutes per mark.
3. The upward green arrow at the end of the green section indicates the official submission deadline for the test.
4. Students must aim to submit their Excel spreadsheet with their answers on the AMS before the official submission deadline.
5. Students can request assistance via **e-mail** ebn2021.ams@gmail.com if they experience any technical issues with respect to uploading their Excel spreadsheets on the AMS.
6. An Upload Grace Period (UGP) of 15 minutes follows the official submission deadline. Students may submit their Excel spreadsheets with answers during this period without incurring any penalties. No email submissions will be accepted during the UGP.
7. A Penalty Period (PP) of 10 minutes follows the UGP. If a student submits an Excel spreadsheets during this period, the student incurs a 10% penalty to their final test mark.
8. No late submissions are accepted after the PP.

Late submissions of session-based assessments, e.g. tests and online practicals



Additional guide for Complex numbers

The following examples illustrate how to appropriately enter complex numbers on the Excel Spreadsheet:

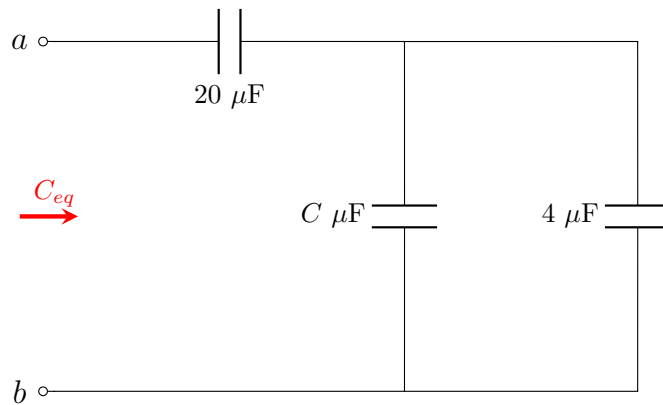
- **Correct:** $1-j2$ or $-3+j4$
 - This is the rectangular form of a complex number. This format is acceptable as a single number entry in the numeric field.
 - Capital J is however incorrect, and $1-2j$ is also not acceptable.
 - If a complex number has no real part, for example $0+j2$, write $0+j2$ as the final answer.
- **Correct:**
 - With angle in degrees: $12\angle 34\deg$;
 - With angle in radians: $12\angle 0.5934$
 - A complex number in its polar form is regarded as a single number entry with the angle in degrees or radians.
- **Incorrect:**
 - $1-i2$ (Use j instead of i)
 - $1 - j2$ (no spaces between characters)
 - $12 \angle 34 \deg$; (no spaces between characters)
 - $j10+1$ (Real part before the imaginary part for rectangular form.)
 - $-3+4j$ (should be $j4$ instead of $4j$)

Section 1

Question 1 [2]

Given: C_{eq}

Calculate the value of C when the equivalent capacitance C_{eq} with respect to terminals $a-b$ is given.



$x = C_{eq}$

01	C [2]
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$$x = 20 \parallel (C+4) = \frac{20(C+4)}{24+C}$$

$$\Rightarrow 24x + Cx = 20C + 80$$

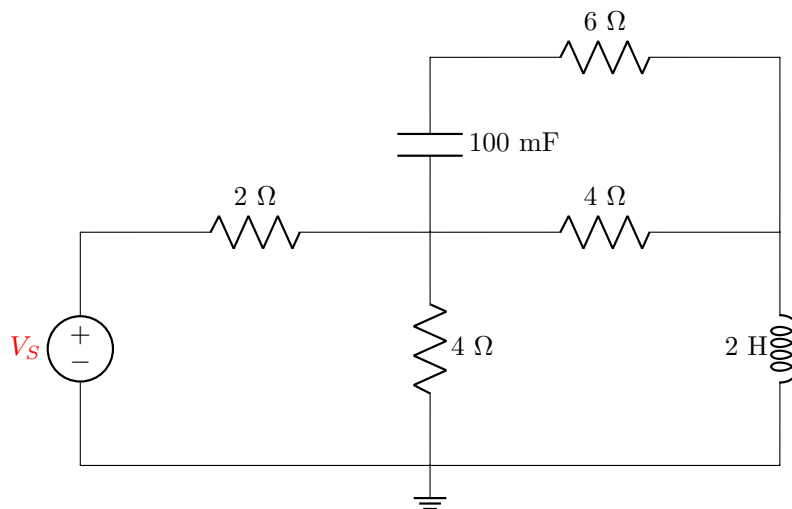
$$\Rightarrow C(x-20) = 80 - 24x$$

$$\textcircled{01} \Rightarrow C = \frac{80 - 24x}{x - 20} \mu\text{F}$$

$$\Rightarrow 4 \leq x < 20, \quad x \text{ is an integer.}$$

Questions 2 to 3 [4]Given: V_S

The circuit below is under steady DC conditions.



$$x = V_S$$

02 Find the energy w_C stored in the capacitor. [2]**03** Find the energy w_L stored in the inductor. [2]

$$V = \frac{1}{2}x, \quad i = \frac{V}{4} = \frac{1}{8}x$$

$$\textcircled{02} \Rightarrow w_C = \frac{1}{2}CV^2 = \frac{1}{2} \times 0.1 \times \frac{1}{4}x^2 = \frac{1}{80}x^2 \text{ J.}$$

$$\textcircled{03} \Rightarrow w_L = \frac{1}{2}Li^2 = \frac{1}{2} \times 2 \times i^2 = \frac{1}{64}x^2 \text{ J.}$$

$$x = 2 : 2 : 20$$

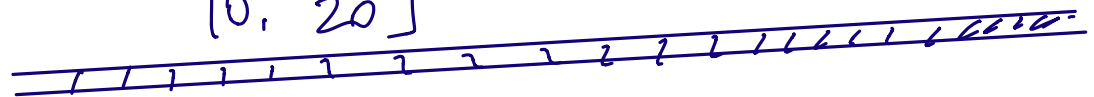
Questions 4 to 5 [4]	
Given: C , L	
The current through a C F capacitor is $i_C(t) = 6 - 3t^2$ A.	
04	Determine the voltage $v_C(t)$ at $t = 2$ s if $v(0) = 2$ V. [2]
The current through an L mH inductor is $i_L(t) = 5e^{-2t} - 2t$ A.	
05	Determine the voltage $v_L(t)$ across the inductor at $t = 3$ s. [2]

$$(04) \quad v_C(t) = \frac{1}{C} \int_0^t (6 - 3t^2) dt + 2$$

$$= \frac{1}{x} \cdot (6t - t^3) \Big|_0^2 + 2$$

$$= \frac{1}{x} (12 - 8) + 2 = \frac{4}{x} + 2 \text{ (V)}$$

$$x = [1, 2, 4, 8, 0.2, 0.5, 0.4, 0.8, 10, 20]$$



$$(05) \quad v_L(t) = L \frac{di(t)}{dt} = y \cdot 10^{-3} (5e^{-2t} - 2t)'$$

$$= 10^{-3} y \cdot [5e^{-2t} \cdot (-2) - 2]$$

$$= 10^{-3} y [-10e^{-2t} - 2]$$

$$v_L(3) = -2.0248 \cdot y \cdot 10^{-3} \text{ J}$$

$$= -2.0248 y \text{ (V)}$$

$$\Rightarrow y = 1:1:10$$

$$\begin{matrix} x = C \\ y = L \end{matrix}$$

Total Section 1: [10]

Section 2

Question 6 [2]

Given: x

Simplify the expression Z . Express the answer in rectangular form .
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$$\mathbf{Z} = \frac{4\angle 90^\circ + (j4 + x)^*}{2e^{j45^\circ} + \sqrt{j4}},$$

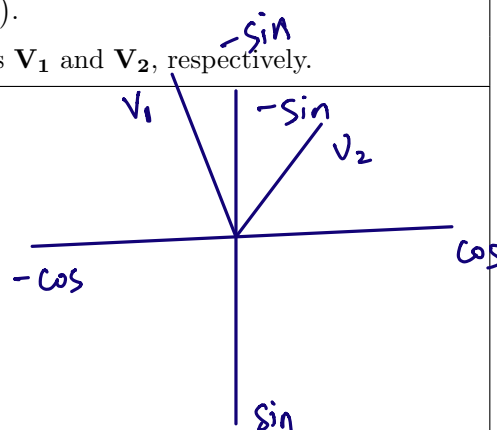
(Note, * refers to the complex conjugate.)
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06	Z (rectangular form) [2]
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$$\begin{aligned} Z &= \frac{j4 + x - j4}{2\angle 45^\circ + 2\angle 45^\circ} = \frac{x}{4\angle 45^\circ} \\ &= \frac{x}{4} \angle -45^\circ. \end{aligned}$$

$$\Rightarrow x = [1, 2, 4, 8, 10, 16, 20, 32, 40, 64]$$

Questions 7 to 8 [4]	
Given: A , B	
Consider sinusoid waveforms $v_1(t)$ and $v_2(t)$, where $v_1(t) = -A \sin(\omega t + 30^\circ) \text{ and}$ $v_2(t) = B \cos(\omega t + 60^\circ).$ The waveforms $v_1(t)$ and $v_2(t)$ can be represented by phasors $\mathbf{V}_1$ and $\mathbf{V}_2$, respectively.	
07	Choose the correct statement. [2] v_2 leads v_1 by 120°. v_1 leads v_2 by 30°. v_1 leads v_2 by 120°. v_2 leads v_1 by 90°. - None of the above is correct.
08	Find $\mathbf{V}_T = \mathbf{V}_1 + \mathbf{V}_2$ in rectangular form. [2]



$$V_1 = A \angle 120^\circ$$

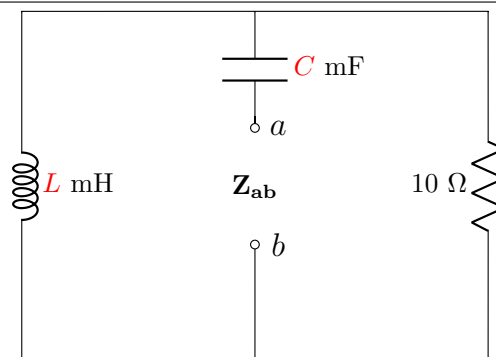
$$V_2 = B \angle 60^\circ$$

$$V_T = A \angle 120^\circ + B \angle 60^\circ$$

$$A = 1:1:10$$

$$B = \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

Question 9 [2]

Given: C , L When $\omega = 100 \text{ rad/s}$, find the equivalent impedance Z_{ab} in **rectangular form** at terminals a - b .

$$\begin{aligned} x &= L \\ y &= C \end{aligned}$$

09 Z_{ab} (rectangular form) [2]

$$Z_L = j 100x \cdot 10^{-3} = j 0.1x$$

$$Z_R = 10$$

$$Z_C = -j \frac{1}{100y \cdot 10^{-3}} = -j \frac{10}{y}$$

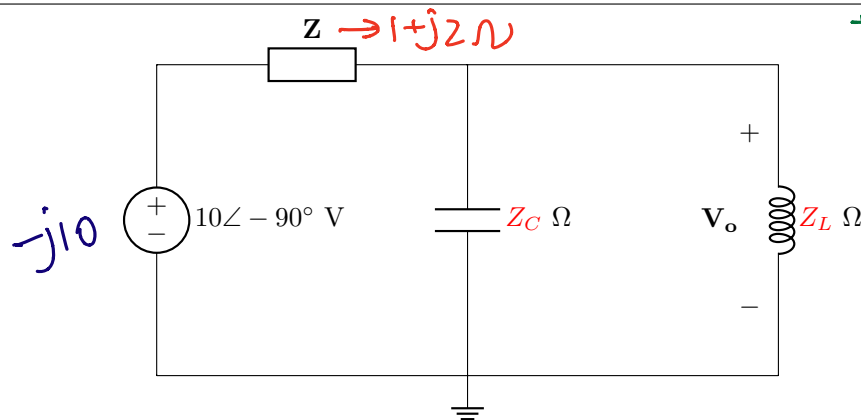
$$\Rightarrow Z_{ab} = 10 \parallel j 0.1x - j \frac{10}{y}$$

$$= \frac{jx}{10 + j 0.1x} - j \frac{10}{y}$$

$$x = 20 \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

$$y = 1:1:10$$

Question 10 [2]

Given: Z_C , Z_L If the impedance $Z = 1 + j2 \Omega$, find V_o in rectangular form.10 V_o (rectangular form) [2]

$$Z_1 = -jX || jY = \frac{XY}{j(Y-X)}$$

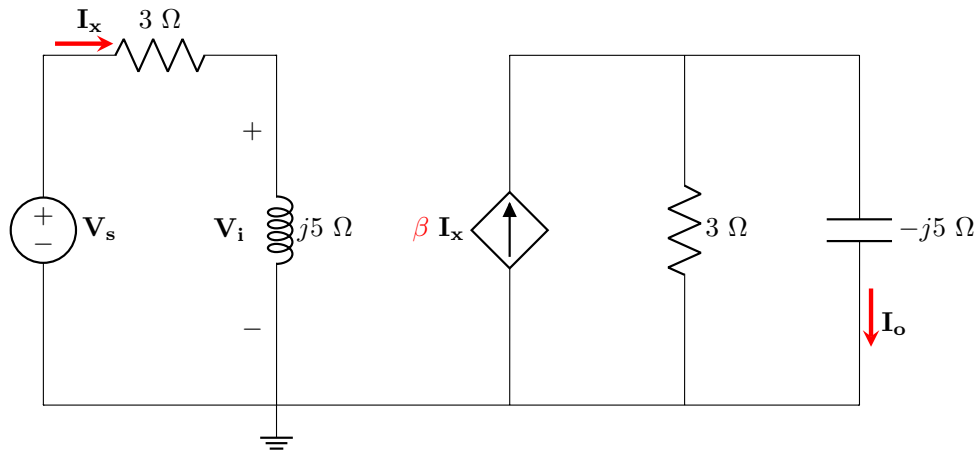
$$V_o = \frac{-j10}{Z + Z_1} = \frac{-j10}{1 + j2 + \frac{XY}{j(Y-X)}}$$

$$= \frac{10(Y-X)}{(-2+j)(Y-X) + XY}$$

$$Y = 2 \begin{bmatrix} 12 & 15 & 24 & 36 & 64 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

$$X = 1: 1: 10$$

Questions 11 to 12 [4]

Given: r, β 11 Given $V_s = r\angle 90^\circ$ V, find the current I_o in rectangular form. [2]12 Given $I_o = 2\angle 90^\circ$ A, find the voltage V_i in rectangular form. [2]

$$\textcircled{11} \quad I_x = \frac{j r}{3 + j 5}, \quad I_o = \beta I_x \cdot \frac{3}{3 - j 5}$$

$$= \beta \cdot \frac{j r}{3 + j 5} \cdot \frac{3}{3 - j 5} = \frac{j 3 \beta r}{34}$$

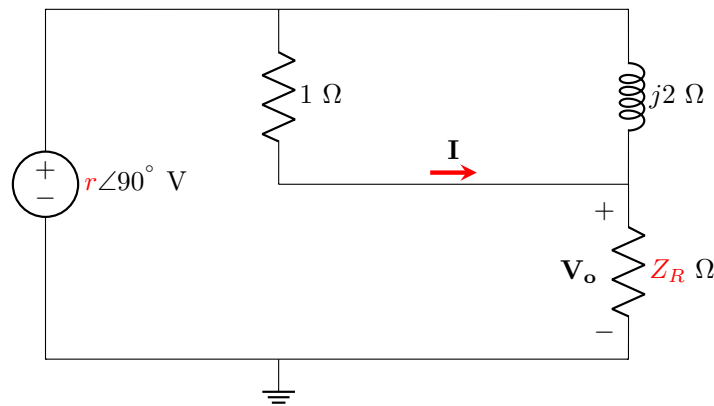
$$\beta = 0.1 : 0.1 : 1, \quad r = 1 : 1 : 10$$

$$\textcircled{12} \quad I_o = \beta I_x \cdot \frac{3}{3 - j 5} = j 2$$

$$\Rightarrow I_x = \frac{j 2 (3 - j 5)}{3 \beta}$$

$$V_i = j 5 I_x = j 5 \cdot \frac{j 2 (3 - j 5)}{3 \beta} = \left(-10 + j \frac{50}{3} \right) / \beta$$

Questions 13 to 14 [4]

Given: r , Z_R 

$$Z_R = X$$

13 Find the voltage V_o in rectangular form. [2]

14 Find the current I in rectangular form. [2]

$$V_o = jr \cdot \frac{X}{X + 1 || j2}, \quad Z = 1 || j2 = \frac{j2}{1 + j2} = 0.8 + j0.4 \, \Omega$$

$$= jr \cdot \frac{X}{X + 0.8 + j0.4}$$

$$I = \frac{V_{1\Omega}}{1\Omega} = jr \cdot \frac{0.8 + j0.4}{X + 0.8 + j0.4}$$

$$r = 20(1:1:10)$$

$$X = \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

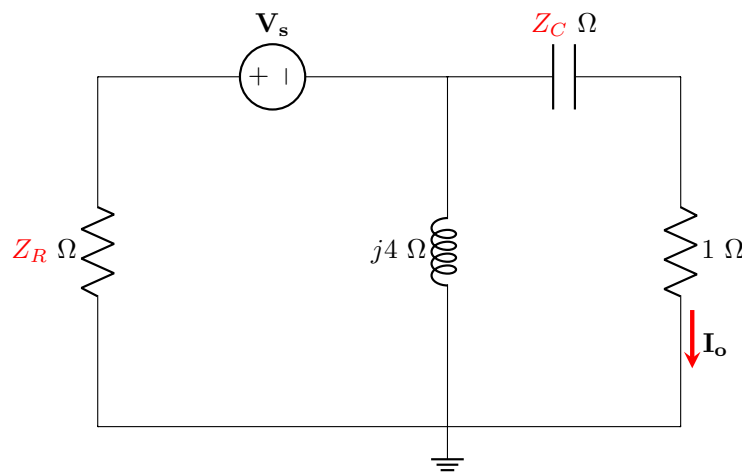
Total Section 2: [18]

Section 3

Question 15 [2]

Given: Z_L , Z_C

When $\mathbf{I}_o = 2\angle 0^\circ$ A, use nodal analysis to find the phasor voltage $\mathbf{V}_s = r\angle\theta$ V in **polar form**, where $r \geq 0$ and $-180^\circ \leq \theta \leq 180^\circ$.



$$Z_R = X$$

$$Z_C = -j4$$

15 $\mathbf{V}_s$ (polar form) [2]

$$V = 2(1 - j4)$$

$$i = \frac{V + V_s}{X} = -2 - \frac{2(1 - j4)}{j4}$$

$$\Rightarrow V_s = -\left[2 + \frac{2(1 - j4)}{j4}\right]X - 2(1 - j4)$$

$$X = 1 : 1 : 10$$

$$Y = \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

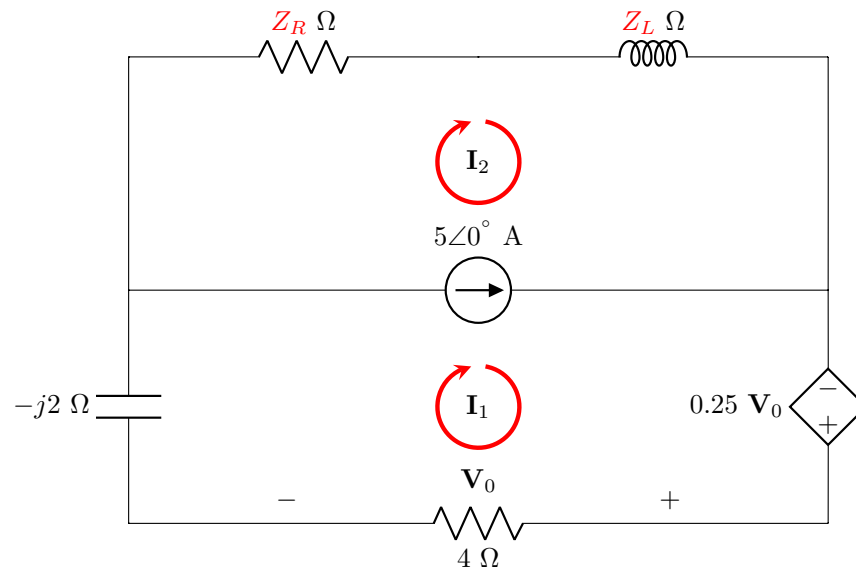
Questions 16 to 19 [4]

Given: Z_R , Z_L

Use mesh analysis to setup two equations of the following form for the circuit below:

$$a\mathbf{I}_1 + b\mathbf{I}_2 = 5\angle 0^\circ,$$

$$(3 - j2)\mathbf{I}_1 + c\mathbf{I}_2 = d.$$

Find the unknown coefficients a , b , c , and d in **rectangular form**.

$$Z_R = x$$

$$Z_L = jy$$

16	a [1]	17	b [1]	18	c [1]	19	d [1]
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$$I_1 - I_2 = 5\angle 0^\circ, \quad V_0 = 4I_1$$

$$(x + jy)I_2 - 0.25V_0 + V_0 - j2I_1 = 0$$

$$(x + jy)I_2 - I_1 + 4I_1 - j2I_1 = 0$$

$$\Rightarrow (3 - j2)I_1 + (x + jy)I_2 = 0$$

$$\Rightarrow a = 1, \quad b = -1$$

$$c = x + jy, \quad d = 0$$

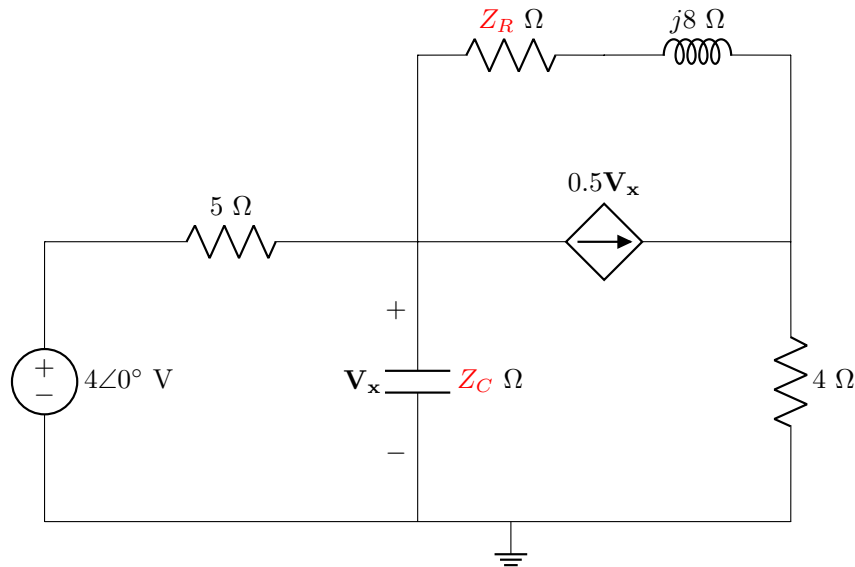
$$x = 1: 1: 10$$

$$y = [1, 2, 4, 6, 8, 10, 12, 16, 20, 25]$$

Questions 20 to 22 [4]

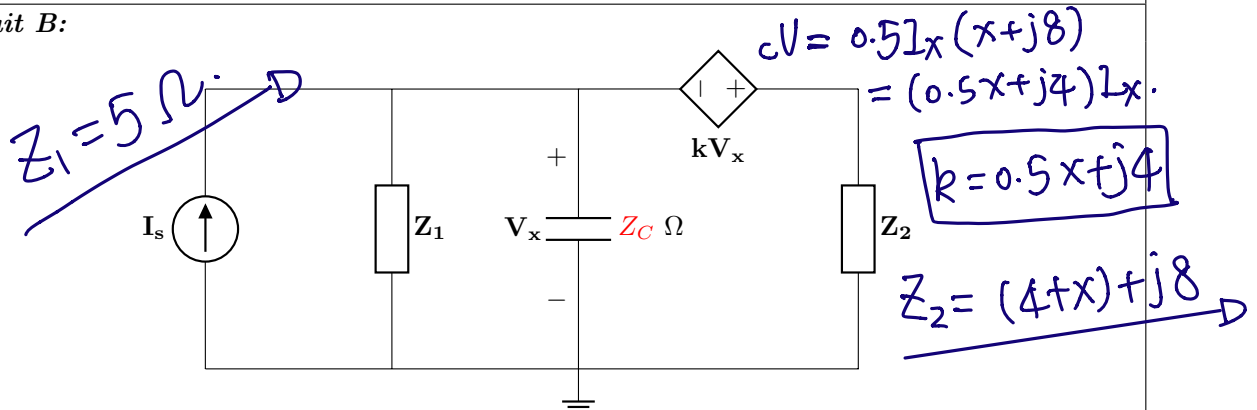
Given: Z_R , Z_C

Use source transformation to transform **Circuit A**, such that it is equivalent to **Circuit B**. In **Circuit B**, find the impedances Z_1 and Z_2 , and the unknown coefficients k .

Circuit A:

$$Z_R = X$$

$$Z_C = -jY$$

Circuit B:

20 In **Circuit B**, find the impedance Z_1 in **rectangular form**. [1]

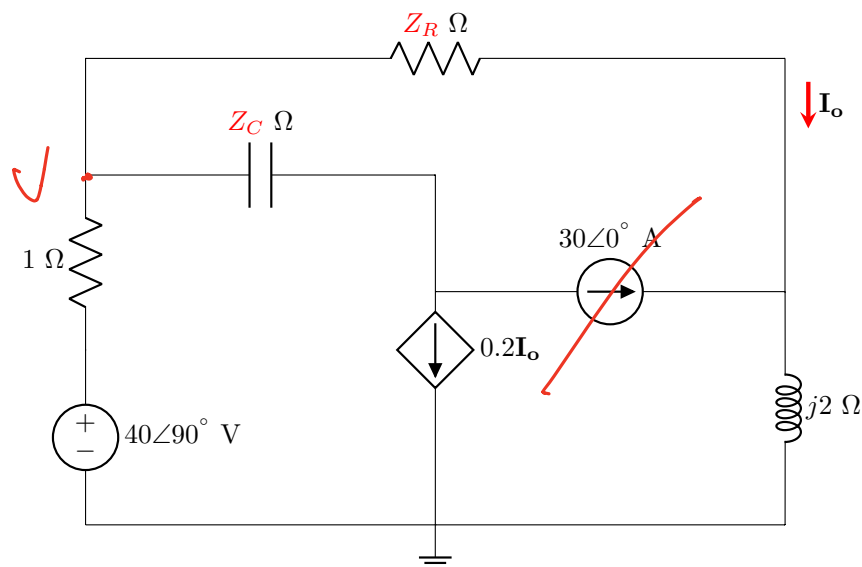
21 In **Circuit B**, find the impedance Z_2 in **rectangular form**. [1]

22 In **Circuit B**, find the coefficient k in **rectangular form**. [2]

$$X = 1:1:10,$$

$$Y = 4 \times \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

Question 23 [2]

Given: Z_R , Z_C Determine $\mathbf{I}_o^V$, the exclusive contribution of the **voltage** source to $\mathbf{I}_o$. Write your answer in **rectangular form**.

$$Z_C = -j4$$

$$Z_R = X$$

23 $\mathbf{I}_o^V$ (rectangular form) [2]

$$\frac{V - j40}{1} + 0.2I_o + I_o = 0$$

$$\Rightarrow V - j40 + 1.2 \cdot \frac{V}{X + j2} = 0$$

$$\Rightarrow \left(1 + \frac{1.2}{X + j2}\right) V = j40$$

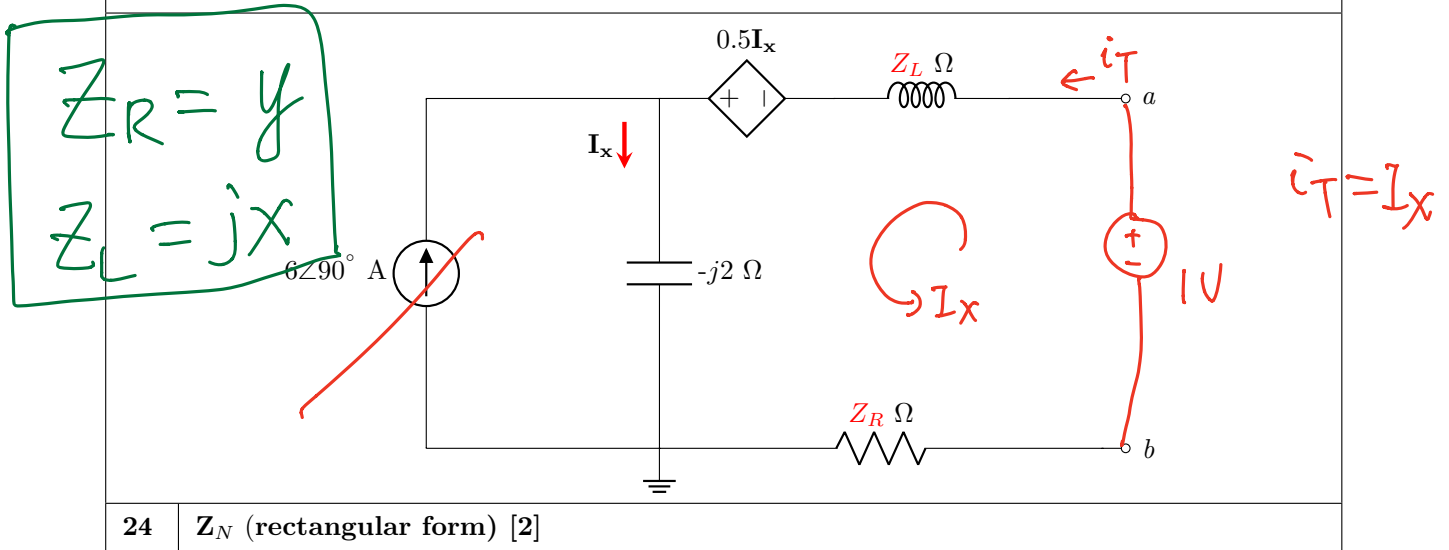
$$\Rightarrow (X + j2 + 1.2) V = j40(X + j2)$$

$$\Rightarrow V = \frac{j40(X + j2)}{X + 1.2 + j2}$$

$$I_o^V = \frac{V}{X + j2} = \frac{j40}{X + 1.2 + j2}$$

$$X = 1:1:10, Y = \begin{bmatrix} 1 & 2 & 4 & 8 & 10 \\ 16 & 20 & 32 & 40 & 50 \end{bmatrix}$$

Question 24 [2]

Given: Z_R , Z_L Find the Norton equivalent impedance Z_N at terminals a - b . Write your answer in **rectangular form**.

$$-1 + [y + j(x-2)]I_x - 0.5I_x = 0$$

$$\Rightarrow [(y - 0.5) + j(x-2)]I_x = 1$$

$$\Rightarrow R_N = \frac{1}{I_x} = y - 0.5 + j(x-2)$$

$$y = 1: 1: 10$$

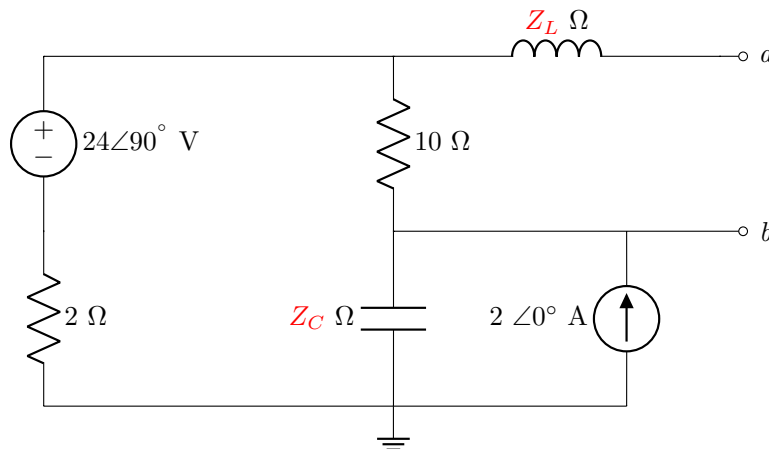
$$x = [1, 2, 4, 8, 10, 12, 16, 18, 20, 24].$$

Questions 25 to 26 [4]

Given: Z_L , Z_C Find the Thévenin equivalent voltage V_{Th} and impedance Z_{Th} at terminals a - b . Write your answers in **rectangular form**.

$$Z_C = jY$$

$$Z_C = -jX$$



$$X = 1 : 1 : 10$$

$$Y = 4 \begin{bmatrix} 1 & 2 & 4 & 8 \\ 10 & 16 & 20 \\ 32 & 40 & 50 \end{bmatrix}$$

25 V_{Th} (rectangular form) [2]26 Z_{Th} (rectangular form) [2]

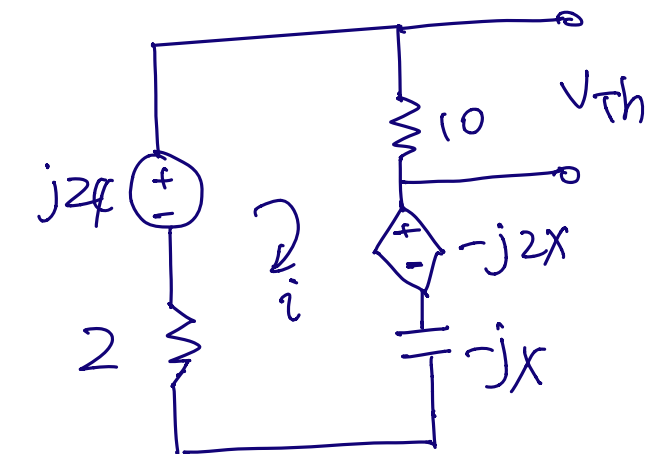
$$Z_{Th} = jY + 10 \parallel (2 - jX) = jY + \frac{10(2 - jX)}{12 - jX}$$

$$V_{Th} = 10i$$

$$\underline{2i} - j24 + \underline{10i} + (-j2X) - \underline{jX \cdot i} = 0$$

$$\Rightarrow (12 - jX)i = j(24 + 2X)$$

$$\Rightarrow i = \frac{j(24 + 2X)}{12 - jX} \Rightarrow$$



$$V_{Th} = 10i = \frac{10j(24 + 2X)}{12 - jX}$$

Total Section 3: [18]

Grand Total : [46]